

Name – Ganesh Malaji Ghule
Roll no - 37

Assignment No 1

- Data Wrangling

localhost:8888/notebooks/College_Practicals/DSBDAL/Practical1.ipynb

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```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

In [2]: df=pd.read_csv('titanic.csv')
df

Out[2]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

891 rows x 12 columns

```
In [3]: df.head()

Out[3]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

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```
In [3]: df.head()

Out[3]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

```
In [4]: df.tail()

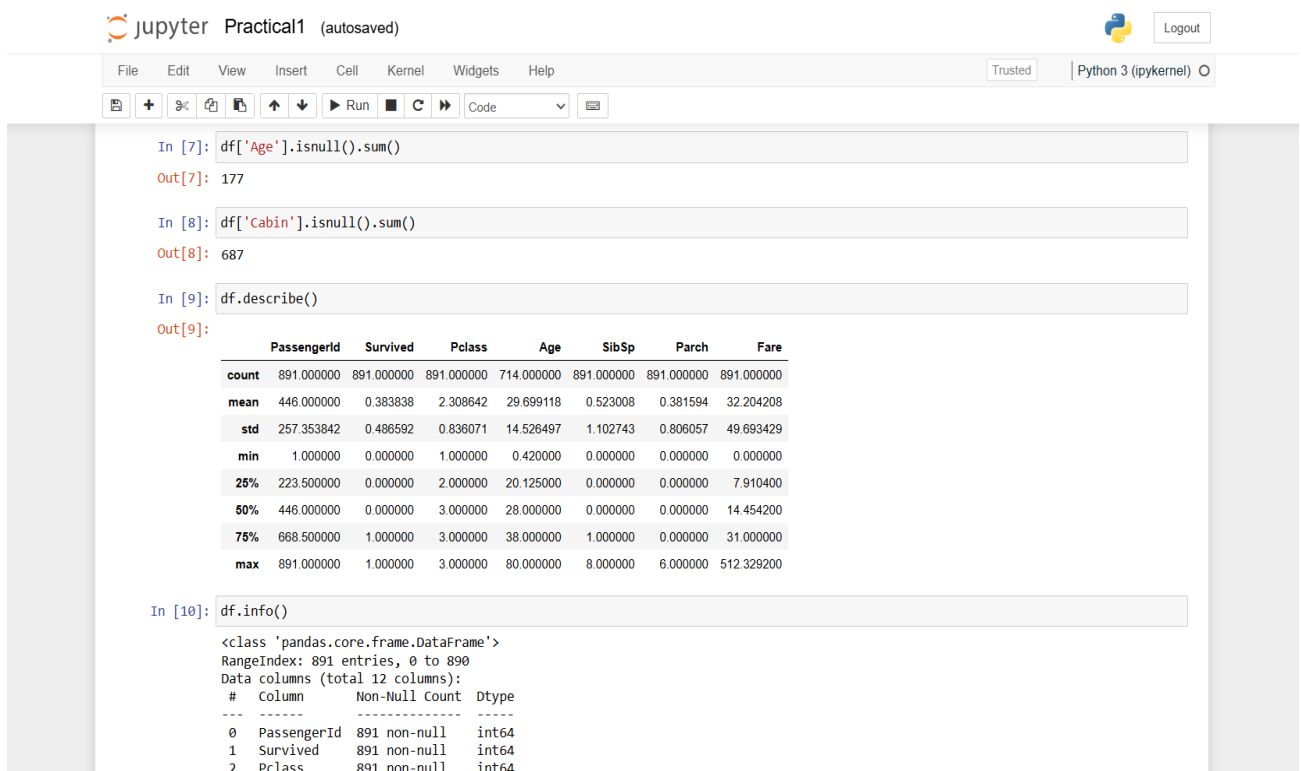
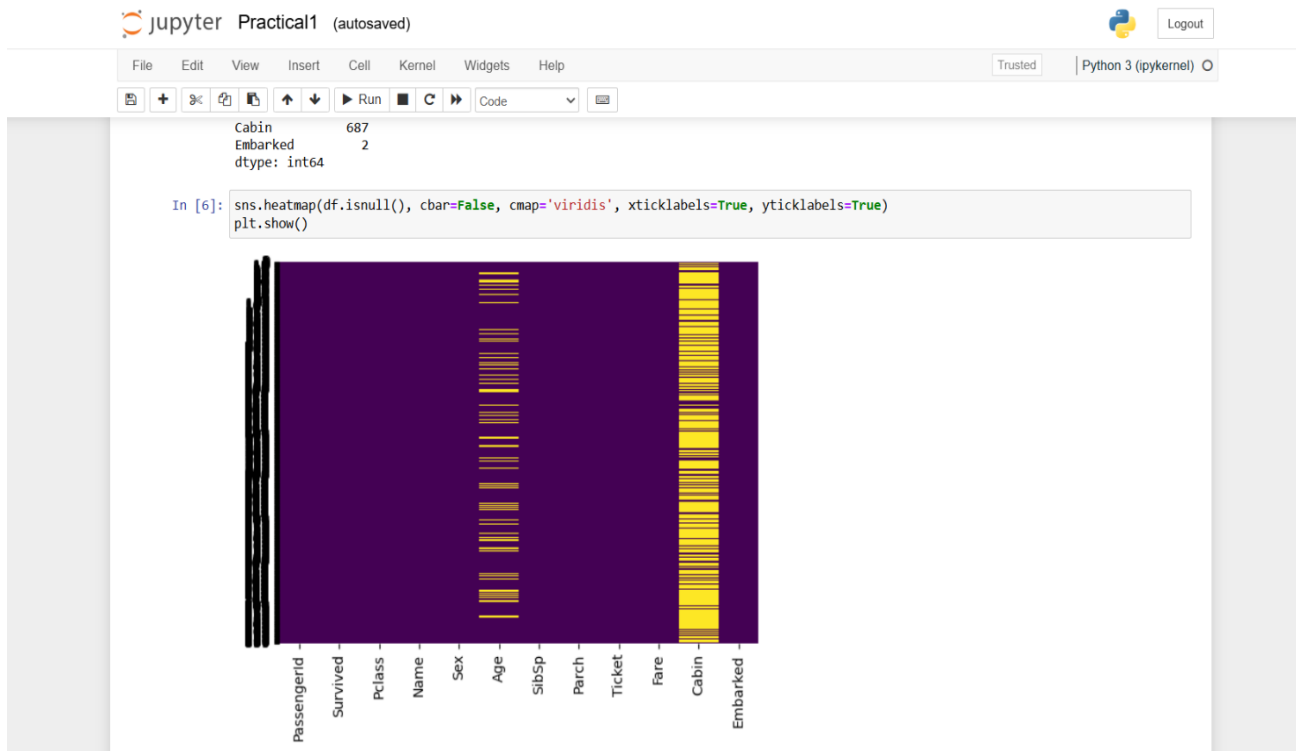
Out[4]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	NaN	Q

```
In [5]: df.isnull().sum()

Out[5]: PassengerId    0
Survived            0
Pclass             0
Name               0
Sex               0
Age              177
SibSp             0
Parch             0
```

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Code

```
0 PassengerId 891 non-null int64
1 Survived 891 non-null int64
2 Pclass 891 non-null int64
3 Name 891 non-null object
4 Sex 891 non-null object
5 Age 714 non-null float64
6 SibSp 891 non-null int64
7 Parch 891 non-null int64
8 Ticket 891 non-null object
9 Fare 891 non-null float64
10 Cabin 204 non-null object
11 Embarked 889 non-null object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [11]: df.dtypes

Out[11]:

PassengerId	int64
Survived	int64
Pclass	int64
Name	object
Sex	object
Age	float64
SibSp	int64
Parch	int64
Ticket	object
Fare	float64
Cabin	object
Embarked	object
dtype:	object

In [12]: df.shape

Out[12]: (891, 12)

In [13]: def impute_age(cols):

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Code

```
In [13]: def impute_age(cols):
Age = cols[0]
Pclass = cols[1]

if pd.isnull(Age):
    if Pclass == 1:
        return 37
    elif Pclass == 2:
        return 29
    else:
        return 24
else:
    return Age
```

In [14]: df['Age'] = df[['Age', 'Pclass']].apply(impute_age, axis=1)
df.drop('Cabin', axis=1, inplace=True)
df.dropna(inplace=True)

In [15]: df.head()

Out[15]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S
1	2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S

In [16]: df.tail()

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      Code 

```
In [16]: df.tail()
```

```
Out[16]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	24.0	1	2	W./C. 6607	23.45	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	Q

```
In [17]: df.isnull().sum()
```

```
Out[17]: PassengerId    0
Survived          0
Pclass            0
Name              0
Sex               0
Age              0
SibSp            0
Parch            0
Ticket           0
Fare             0
Embarked         0
dtype: int64
```

```
In [18]: df['Age'] = df['Age'].round(0).astype('int')
```

```
In [19]: df.dtypes
```

```
Out[19]: PassengerId    int64
Survived      int64
Pclass        int64
```

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      Code 

```
Age          int64
SibSp        int64
Parch        int64
Ticket       object
Fare         float64
Embarked     object
dtype: object
```

```
In [20]: df.head()
```

```
Out[20]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.2500	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38	1	0	PC 17599	71.2833	C
2	3	1	3	Heikinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.9250	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1000	S
4	5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.0500	S

```
In [21]: cat = pd.get_dummies(df, columns=['Sex'])
cat.head()
```

```
Out[21]:
```

	PassengerId	Survived	Pclass	Name	Age	SibSp	Parch	Ticket	Fare	Embarked	Sex_female	Sex_male
0	1	0	3	Braund, Mr. Owen Harris	22	1	0	A/5 21171	7.2500	S	0	1
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	38	1	0	PC 17599	71.2833	C	1	0
2	3	1	3	Heikinen, Miss. Laina	26	0	0	STON/O2. 3101282	7.9250	S	1	0
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	35	1	0	113803	53.1000	S	1	0
4	5	0	3	Allen, Mr. William Henry	35	0	0	373450	8.0500	S	0	1

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3	4	1	1	Futrell, Mrs. Jacques Heath (Lily May Peel)	35	1	0	113803	53.1000	S	1	0
4	5	0	3	Allen, Mr. William Henry	35	0	0	373450	8.0500	S	0	1

```
In [22]: cat['Sex_female']
Out[22]: 0      0
          1      1
          2      1
          3      1
          4      0
          ..
          886    0
          887    1
          888    1
          889    0
          890    0
          Name: Sex_female, Length: 889, dtype: uint8

In [23]: cat['Sex_male']
Out[23]: 0      1
          1      0
          2      0
          3      0
          4      1
          ..
          886    1
          887    0
          888    0
          889    1
          890    1
          Name: Sex_male, Length: 889, dtype: uint8

In [ ]:
```